DATATHON HRT · JUNE 2026

Financial Sentiment & Sharpe Optimization Pipeline

High-performance trading signal pipeline combining price-action features with granular financial sentiment analysis

Team 6

THE CHALLENGE

Predicting Simulated Market Returns

The Problem

Given OHLC bars and news headlines, predict the next-period return ***R*** for each session.

Key constraints:

- High-noise environment with limited signal
- Sharpe ratio as the optimization target (scale-invariant)
- Must generalize across seen/unseen sessions
- Headline text requires NLP processing

Our Approach

Price-Action

19+ OHLC features
(momentum, volatility, risk)

NLP Sentiment

FinBERT + LLM pipeline
for headline scoring

Sharpe Optimizer

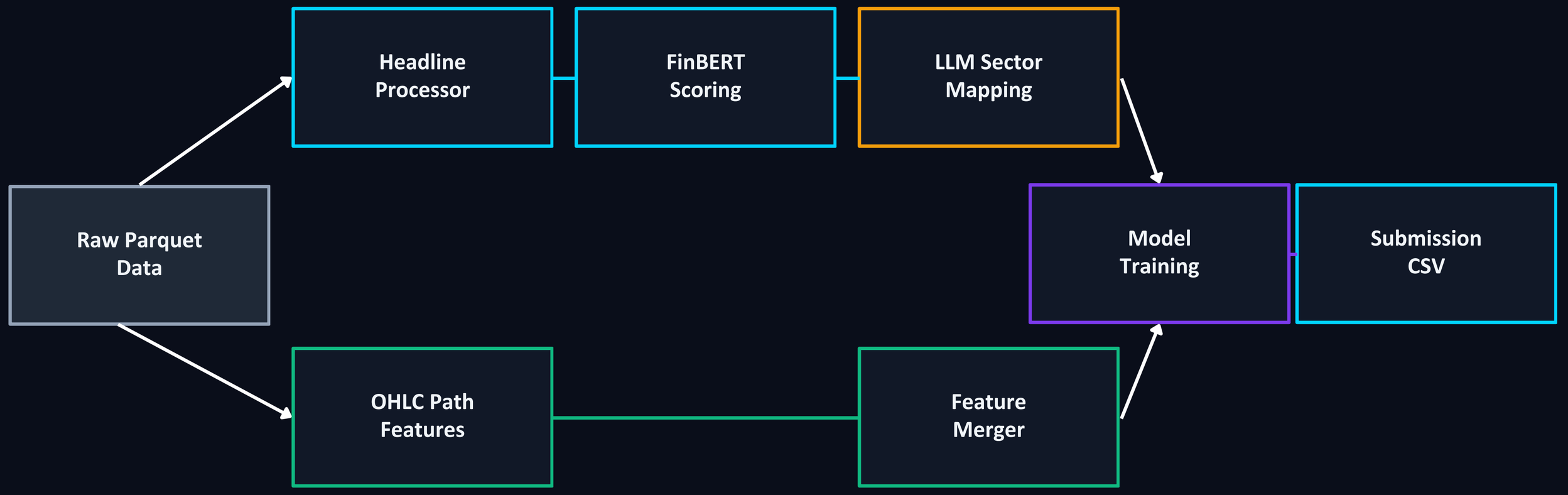
Ridge warm-start + SLSQP
regularized objective

Regime Detection

Gaussian HMM for latent
market state modeling

SYSTEM DESIGN

Pipeline Architecture



Sentiment path

Price-action path

Optimization path

FEATURE ENGINEERING

19+ OHLC & Sentiment Features

Price-Action (OHLC)

Momentum

Cumulative returns · last-5-bar returns · log-close slope

Volatility

Realized vol · Parkinson variance · regime splits (1H vs 2H)

Risk

Max drawdown · bar-return skewness · p95 extreme moves

Trend Stability

R^2 of log-price linear regression

Headline Sentiment

`sentiment_ret_corr`

Correlation between bar returns and headline sentiment scores

`sentiment_weighted_mean`

Sentiment score weighted by FinBERT confidence

`sector_entropy`

Diversity of sectors mentioned within a session

`entity_score`

Primary company signal from word-0 entity extraction

NLP & LLM INTEGRATION

Sentiment Pipeline: FinBERT + LLM

01

Initialization

get_or_process_file entry point called — checks cache first to avoid redundant API calls

02

Local Scoring

FinBERT (ProsusAI/finbert) generates P_pos − P_neg score ∈ [−1, 1]; accelerated on MPS/CUDA

03

LLM Analysis

Batches sent to Gemini or Ollama for company extraction + reasoning per headline

04

Sector Mapping

Unique granular sectors mapped to 5 canonical categories in a single LLM pass

05

Caching

Results saved as {filename}_processed_{model}.parquet to prevent redundant API calls

Gemini Primary cloud LLM · **Ollama / llama3** 100% local inference · **MPS / CUDA** GPU acceleration for FinBERT

MODELING STRATEGY

Sharpe Optimization & Ensemble

Primary: Sharpe-Augmented Linear Ensemble

Anchors weights to a stable Ridge model to prevent overfitting in high-noise financial environments.

OBJECTIVE FUNCTION

$$J(\beta) = -\text{Sharpe}((X\beta) \cdot R) + \lambda \cdot \text{MSE}(X\beta, w_{ridge})$$

Ridge Warm-Start → SLSQP

Ridge provides stable initial weights; SLSQP optimizer then maximizes Sharpe ratio directly via constrained gradient descent.

Gaussian HMM Regime Detection

Hidden Markov Model identifies latent market regimes (trending vs. mean-reverting). Linear head fitted on top of posterior state probabilities.

Distributional Monotonicity

Uses prob_sign or quantile_median to determine position direction — focuses on sign prediction rather than magnitude.

Sharpe ratio is scale-invariant → final weights normalized for consistent risk profile across submissions.

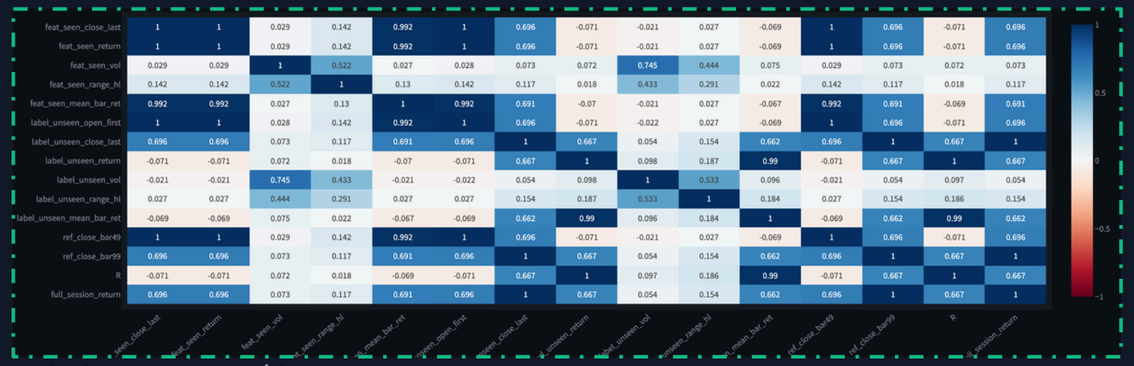
INTERACTIVE EXPLORATION

Data Visualizations & Insights

Candlestick + News Events

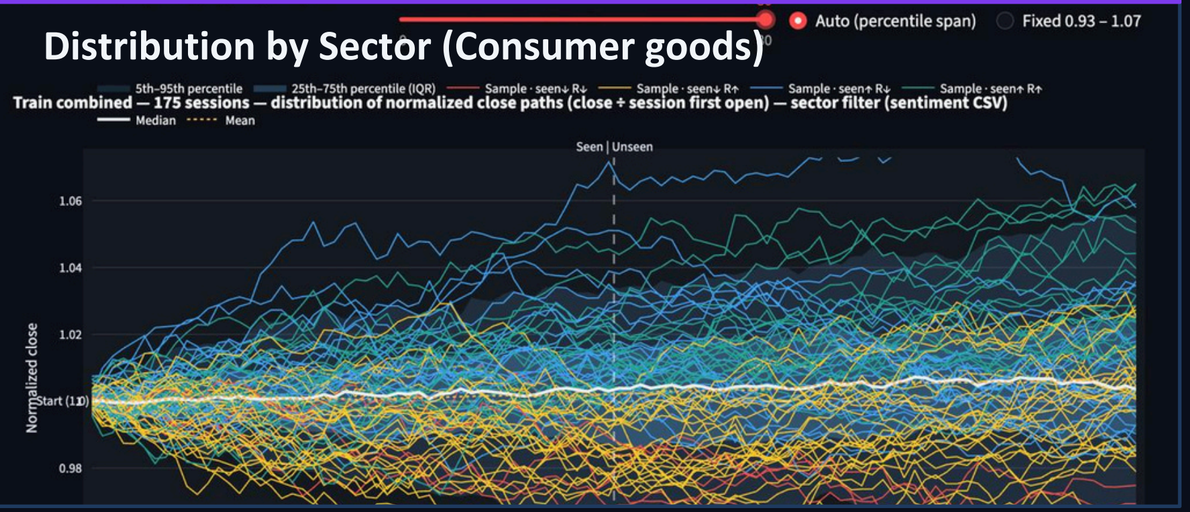
Session Explorer

Feature Correlation Matrix

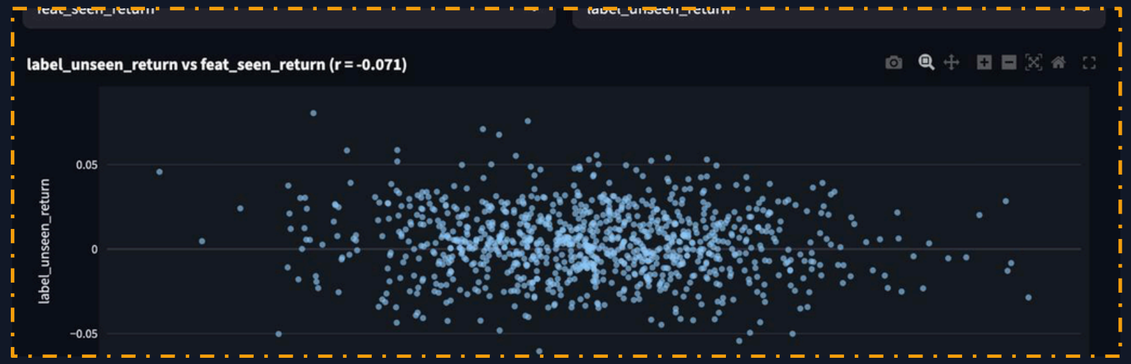


Train Set · Pearson / Spearman

Distribution by Sector (Consumer goods)



Scatter Everywhere



SUMMARY

Key Takeaways

Multi-modal signals win

Combining price-action OHLC features with granular FinBERT sentiment captures information that neither source alone provides.

Sharpe > MSE as objective

Directly optimizing for risk-adjusted returns (with Ridge regularization) outperforms naive MSE minimization in noisy markets.

LLM orchestration adds structure

Gemini/Ollama-powered entity & sector extraction enforces cross-session consistency that bag-of-words approaches miss.

Hardware-aware inference

MPS/CUDA acceleration + caching makes the full sentiment pipeline practical for datathon time constraints.

SUMMARY

Team

Jonas

The Executioner

- B.Sc. ETH ITET
- The Finance Guy
- YAY :)

Arka

The Viber

- Ex-Microsoft
- MSc ETH
- BTech IIT Kharagpur

Shubham Chowdhary

The Thinker

- MSc ETH
- Ex-HFT
- BTech IIT Roorkee